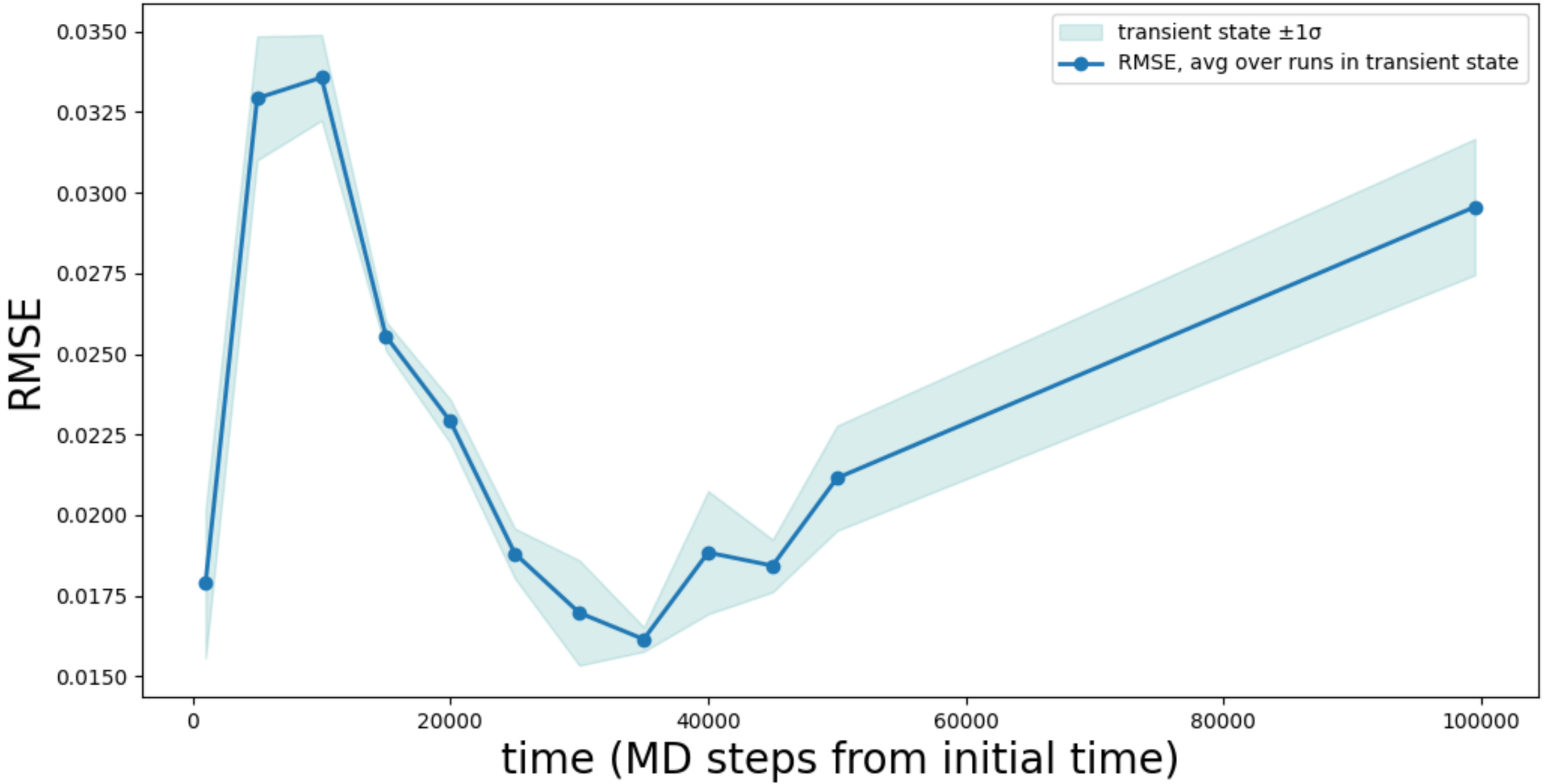
transient state ±1σ
RMSE, avg over runs in transient state
RMSE
0.0350
0.0325
0.0300
0.0275
0.0250
0.0225
0.0200
0.0175
0.0150
0
20000
40000
60000
80000
100000
time (MD steps from initial time)

Transient era result:

$$RMSE_{\Delta x}(\Delta t)$$

10

$\Delta x \approx 0 \rightarrow$

low error

small portion of box with $\Delta x > 0$, but inadequate training data

Steady drop for $\Delta t \geq 15000$: shear information propagates to greater portion of particles

t_c

Steady rise for $\Delta t \geq t_c \approx 35000$: behavior is more complex and less correlated to wall distance

1 MD step $=$ strain γ of 10^{-8}

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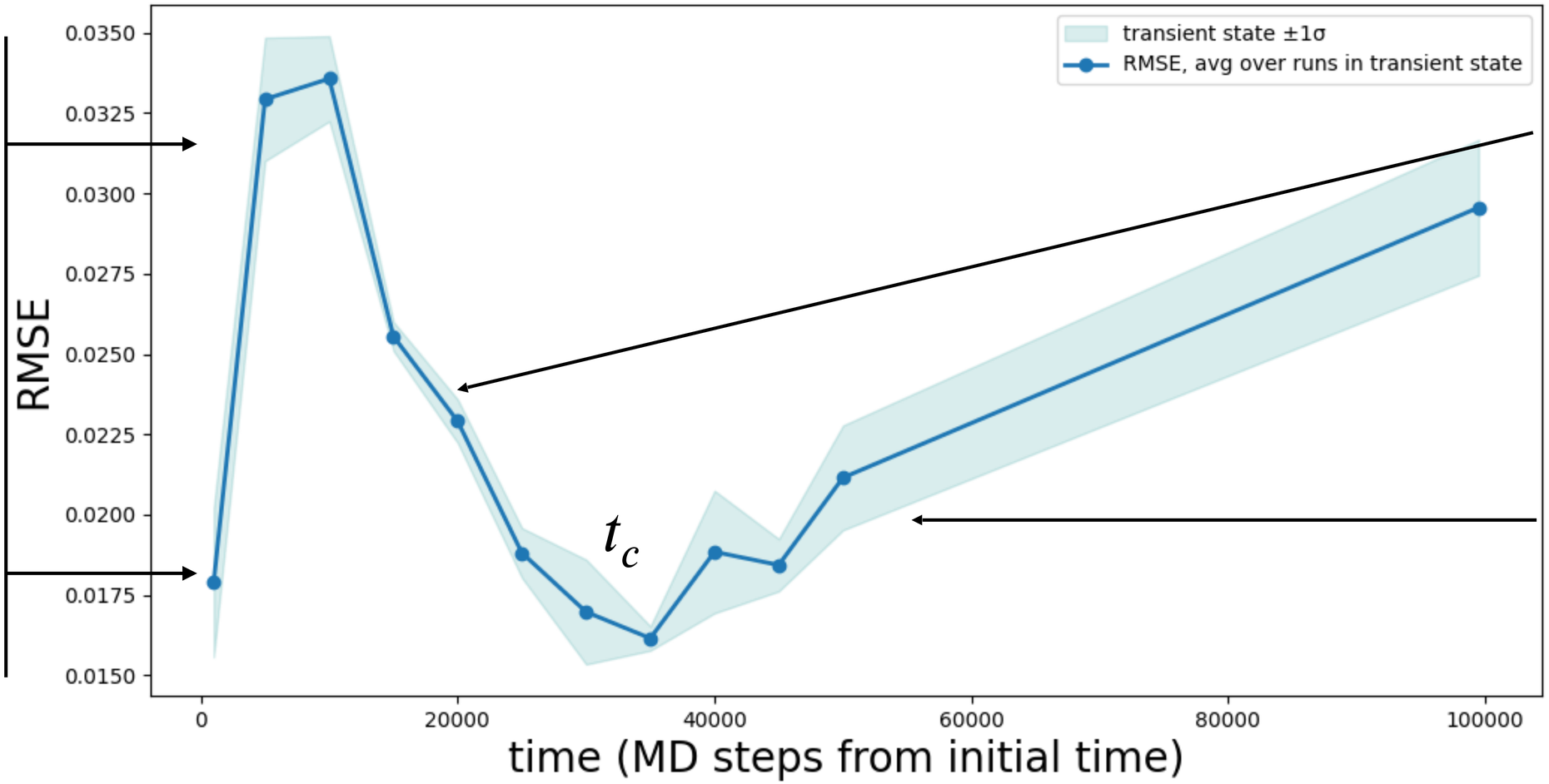

Steady drop for $\Delta t \geq 15000$: shear information propagates to greater portion of particles

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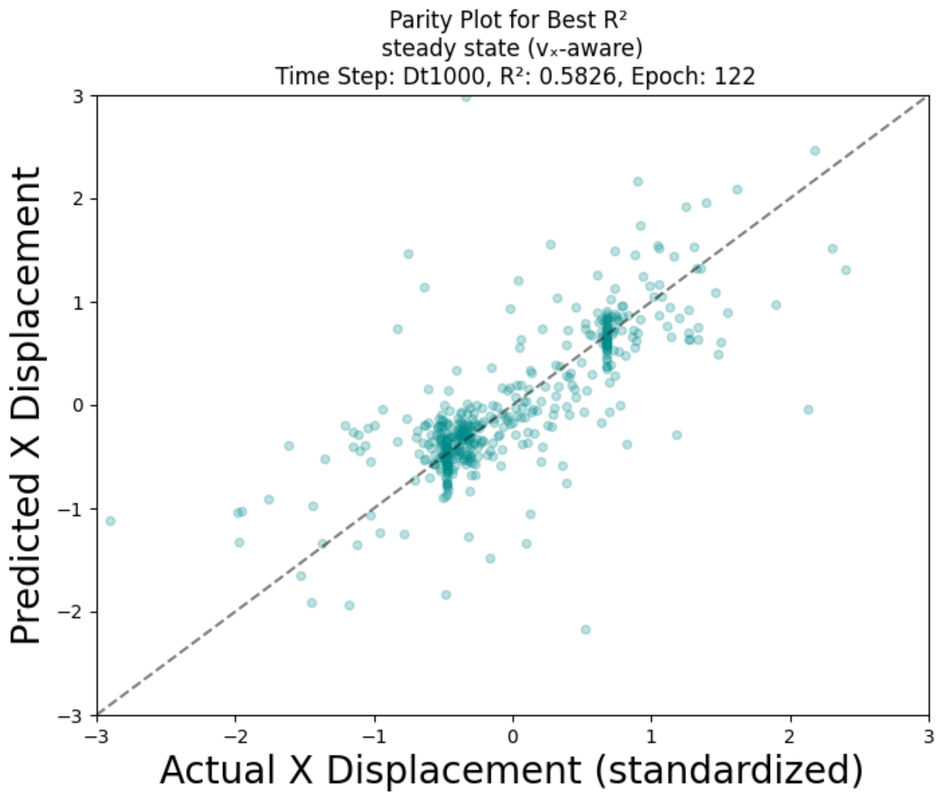
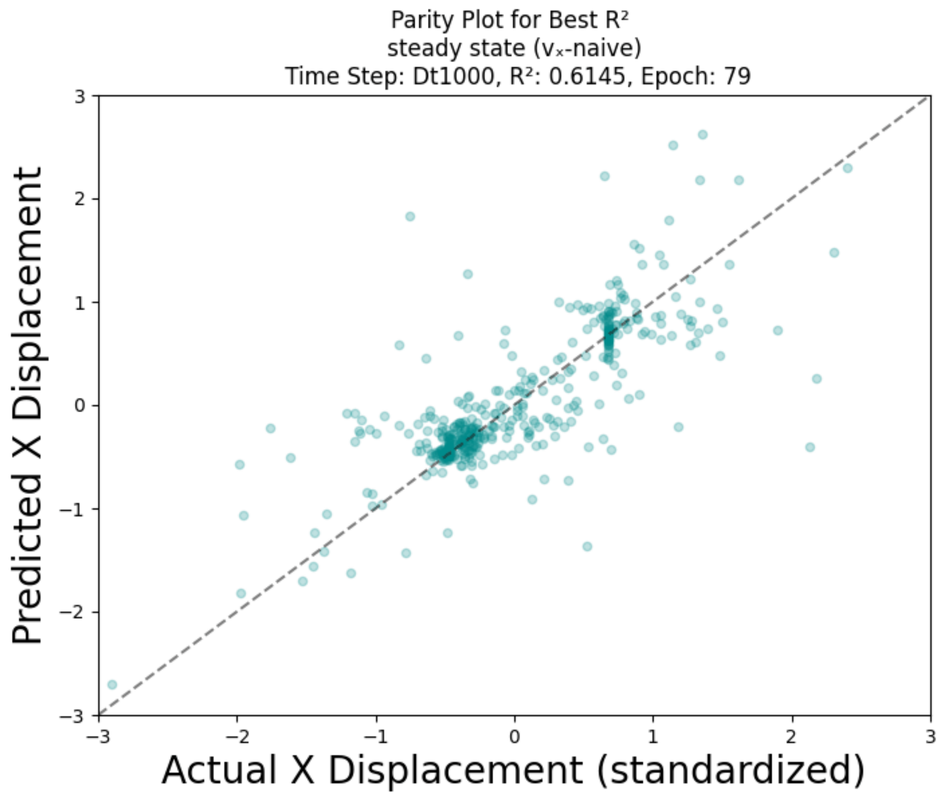
Steady state era result:

worsened performance, even with $v_x(t = 0)$ as input